Prior-Guided Flow Matching for Missing Modality MRI Synthesis

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***Abstract*—Multi-modal MRI provides complementary diagnostic information, but complete sequence sets are often unavailable. We present PriorFM, a prior-guided flow-matching framework that synthesizes a specified target modality from arbitrary subsets of available modalities using one task-conditioned model per dataset. Conditioned on the target identity, source mask, availability, and source count, PriorFM fuses observed modalities into a coarse anatomical prior. A conditional flow network refines this prior under anatomical and frequency-domain guidance, enabling deterministic synthesis with few Euler steps. Experiments on BraTS2021, CHAOS, and Diff5T cover 32 source-to-target tasks and show the best average synthesis performance among the evaluated methods. The synthesized images also support downstream segmentation, with Dice scores close to those obtained from real target scans. Component and sampling-step analyses demonstrate the contributions of task-aware fusion, frequency cues, and prior-context conditioning. Code and pretrained weights are available at https://github.com/hekaiwang/PriorFM.**

***Index Terms*—MRI synthesis, missing modality, flow matching, multi-modal fusion, image generation.**

I. Introduction

Multi-modal magnetic resonance imaging (MRI) provides complementary anatomical and pathological information for clinical diagnosis and image analysis. Complete acquisitions are often unavailable because of limited scan time, patient motion, contrast-agent contraindications, or heterogeneous protocols. Missing modalities can consequently degrade downstream tasks such as segmentation [1].

Missing-modality synthesis recovers unavailable sequences from observed images [2]–[7]. GAN-based [8]–[13] and diffusion-based methods [14]–[19] have advanced cross-modality translation, while multi-source approaches [4], [20]–[24] exploit complementary inputs. Many methods nevertheless use predefined source configurations or fixed targets. As illustrated in Fig. 1, arbitrary missing-modality synthesis requires a unified model that accommodates varying source subsets and target modalities [5]–[7], [16], [22].

This setting requires target-aware aggregation of variable-size source sets [25], [26] and preservation of anatomical structure during modality-specific appearance generation. Different source combinations provide unequal information for the same target, while the same source set may serve different synthesis tasks. The observed images already contain anatomical information that can guide generation from a structured initialization.

We present PriorFM, which formulates synthesis as task-conditioned prior-to-target transport. The synthesis task encodes the target identity, source and availability masks, and source count, allowing the same network to adapt its fusion and refinement to each configuration. A task-aware fusion encoder aggregates observed modalities into a target-specific coarse prior and multi-scale contextual representations. Conditional flow matching [18], [27] then refines the prior through a learned velocity field, guided by task information, prior context, and Haar wavelet frequency cues [28]. Starting from this anatomical prior enables deterministic synthesis with few Euler steps. One model is trained per dataset and shared across its source–target configurations. Our contributions are:

- A task-conditioned formulation and masked multimodal fusion strategy that construct a target-specific anatomical prior from arbitrary observed source subsets.
- A conditional flow refinement scheme that combines prior-context and frequency guidance along a short deterministic prior-to-target trajectory.
- Evaluation across 32 tasks on BraTS2021 [29], CHAOS [30], and Diff5T [31], demonstrating improved synthesis quality and downstream segmentation utility with few sampling steps.

II. Method

A. Unified Task Formulation and Sampling

Let $\mathcal{M} = \{m_1, \ldots, m_K\}$ be a fixed modality registry and $\mathcal{A} \subseteq \mathcal{M}$ the acquired modalities of a training subject. Given a target $m_t \in \mathcal{A}$, we synthesize x_{m_t} from a non-empty source subset $\mathcal{S} \subseteq \mathcal{A} \setminus \{m_t\}$. The target image is used only for supervision. The unified input $\tilde{\mathbf{x}} = \{\tilde{x}_m\}_{m\in\mathcal{M}}$ follows the registry order, with unselected and unavailable modalities zero-filled:

$$\tilde{x}_m = \begin{cases} x_m, & m \in \mathcal{S}, \\ 0, & \text{otherwise.} \end{cases} \tag{1}$$

The source and availability masks $\mathbf{s}, \mathbf{a} \in \{0,1\}^K$ have entries $s_k = \mathbb{I}(m_k \in \mathcal{S})$ and $a_k = \mathbb{I}(m_k \in \mathcal{A})$, respectively. PriorFM

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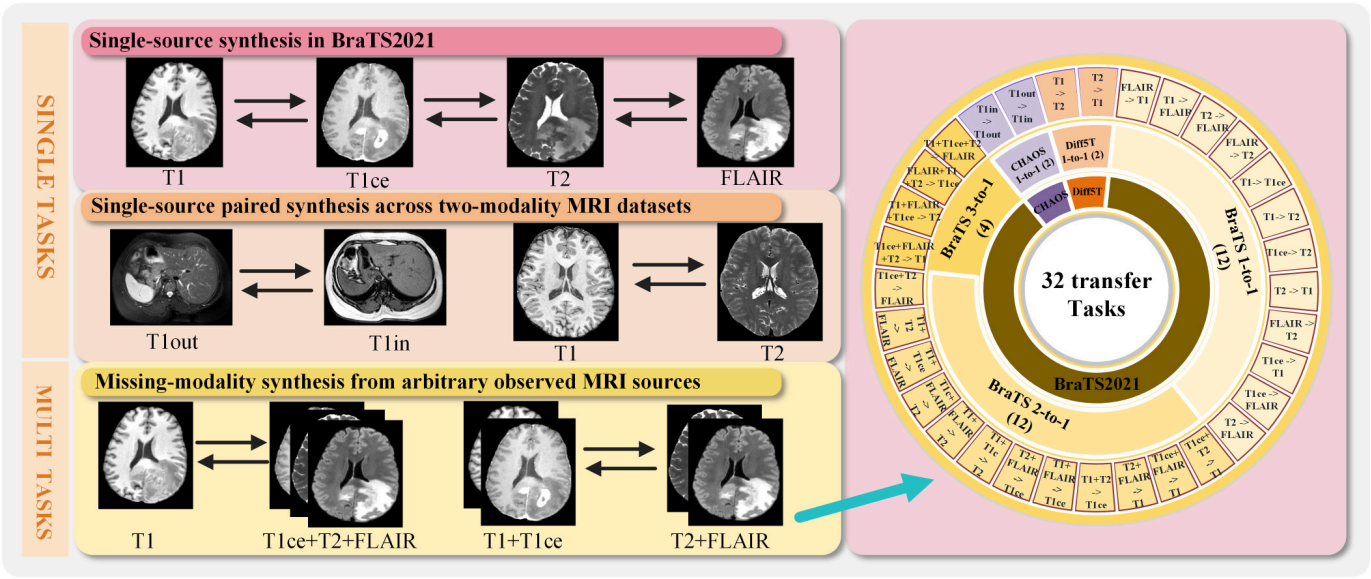


Fig. 1. Unified MRI synthesis across 32 source–target tasks: 28 on BraTS2021 and two each on CHAOS and Diff5T.

learns $\hat{x}_{m_t} = f_\theta(\tilde{\mathbf{x}}, \mathbf{s}, \mathbf{a}, m_t)$, where $\hat{x}_{m_t}$ is the synthesized target.

Targets and source subsets are dynamically sampled during training, covering single-source and multi-source synthesis with the same model. For the four BraTS2021 modalities, this includes all single-source, two-source-to-one, and three-source-to-one tasks. Many-to-many completion queries different targets while keeping the observed source set fixed. We use a 2.5D representation: adjacent slices provide input context, and the target center slice provides supervision. This preserves local through-plane anatomical context while retaining the computational efficiency of 2D synthesis.

B. PriorFM Architecture

As shown in Fig. 2, PriorFM combines task-aware multimodal fusion, coarse prior generation, a Haar wavelet adapter, and conditional flow refinement. A shared encoder E_ϕ extracts L feature scales from each modality, while a task encoder combines the masks, target embedding $\mathbf{e}_{m_t}$, and source-set statistics $\mathbf{r}_\mathcal{S}$, including the source count:

$$\begin{aligned} \{\mathbf{F}^\ell_{m_k}\}_{\ell=1}^L &= E_\phi(\tilde{x}_{m_k}), \\ \mathbf{c}_t &= E_{\text{task}}(\mathbf{s}, \mathbf{a}, \mathbf{e}_{m_t}, \mathbf{r}_\mathcal{S}), \\ \mathbf{H}^\ell &= \text{Fuse}_\ell(\{\mathbf{F}^\ell_{m_k}, \mathbf{e}_{m_k}, \mathbf{c}_t \mid s_k = 1\}). \end{aligned} \tag{2}$$

Here, $\mathbf{e}_{m_k}$ is the source-modality embedding. Although the encoder also processes zero-filled inputs, masked fusion aggregates only selected sources into the target-aware pyramid $\mathbf{H} = \{\mathbf{H}^\ell\}_{\ell=1}^L$ and compact token $\mathbf{S}$. The task embedding identifies both the observed modalities and the requested target. Compared with direct channel concatenation, masked set fusion accommodates variable-size source sets and target-dependent modality contributions.

The generator G_ψ predicts a coarse anatomical prior p_0, multi-scale context $\mathbf{C}$, and global token $\mathbf{g}_0$. The Haar adapter W_η extracts low- and high-frequency cues from p_0:

$$\begin{aligned} p_0, \mathbf{C}, \mathbf{g}_0 &= G_\psi(\mathbf{H}, \mathbf{S}, \mathbf{c}_t), \\ \mathbf{q}_{\text{low}}, \mathbf{q}_{\text{high}} &= W_\eta(p_0), \\ \Gamma_t &= (\mathbf{c}_t, \mathbf{S}, \mathbf{C}, \mathbf{g}_0, \mathbf{q}_{\text{low}}, \mathbf{q}_{\text{high}}). \end{aligned} \tag{3}$$

The prior p_0 and the conditioning set Γ_t play complementary roles. The prior provides the image-space initialization, while Γ_t supplies task-specific information throughout refinement. In particular, $\mathbf{C}$ provides anatomical context at multiple spatial scales, and $\mathbf{g}_0$ summarizes global structure. The low-frequency cues describe coarse intensity patterns, whereas the high-frequency cues represent rapid spatial variations associated with boundaries and fine detail. Together with $\mathbf{c}_t$ and $\mathbf{S}$, these representations condition the velocity prediction on the requested contrast and the observed source anatomy. For a fixed source–target task, Γ_t remains unchanged during inference, while the current image state evolves through successive Euler updates.

C. Prior-Guided Conditional Flow Matching

We learn a flow from the source-conditioned prior to the target image. For training, the prior is detached as $\bar{p}_0 = \text{sg}(p_0)$, where $\text{sg}(\cdot)$ denotes stop-gradient. At flow time $\lambda \sim \mathcal{U}(\epsilon, 1-\epsilon)$, the linear path and its target velocity are

$$\begin{aligned} z_\lambda &= (1-\lambda)\bar{p}_0 + \lambda x_{m_t}, \\ u^* &= x_{m_t} - \bar{p}_0. \end{aligned} \tag{4}$$

The intermediate state z_λ interpolates between the detached prior and ground-truth target, with u^* specifying the corresponding transport direction. The network predicts $\hat{u} =$

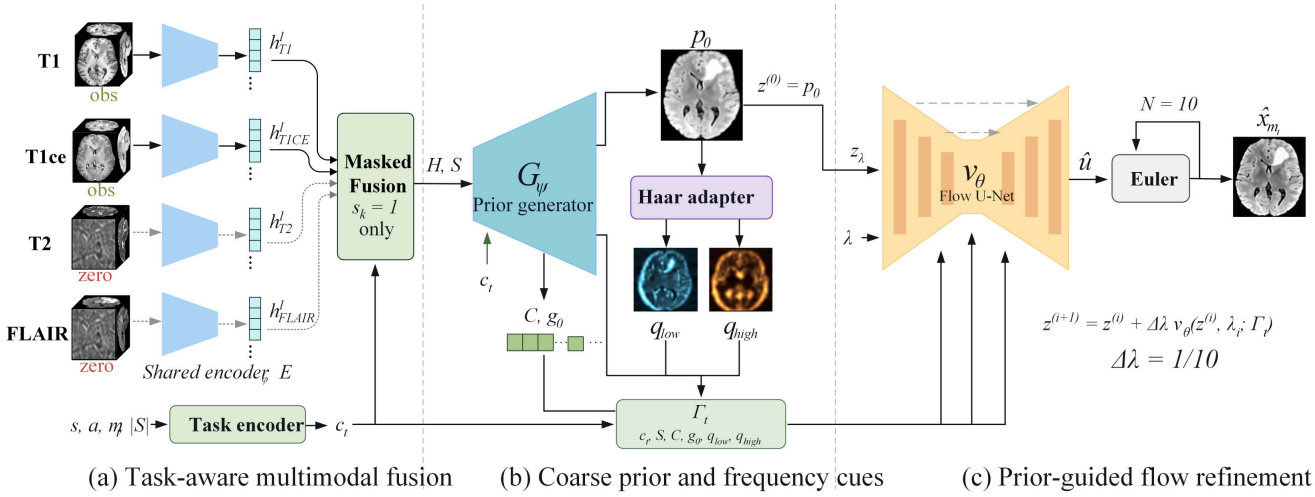


Fig. 2. PriorFM: task-aware fusion and prior generation produce an anatomical prior p_0, multi-scale context $\mathbf{C}$, and global token $\mathbf{g}_0$. The Haar adapter extracts frequency cues from p_0, and conditional flow refinement updates $z^{(i)}$ at each Euler step. Zero labels indicate zero-filled inputs.

$v_\theta(z_\lambda, \lambda; \Gamma_t)$ from the intermediate image, time, and conditioning set. Its flow-matching objective is

$$\mathcal{L}_{\mathrm{FM}} = \mathbb{E}_\lambda \left[\|\hat{u} - u^*\|_2^2 \right]. \tag{5}$$

Starting from an anatomical prior allows the velocity field to model the residual transport toward the target modality. The refinement network thus focuses on modality-specific appearance and fine anatomical detail along a short prior-to-target trajectory.

D. Training Objective

We combine flow matching with target and prior reconstruction, frequency supervision, and a refinement margin:

$$\begin{aligned} \mathcal{L} = \lambda_{\mathrm{FM}}\mathcal{L}_{\mathrm{FM}} + \lambda_{\mathrm{rec}}\mathcal{L}_{\mathrm{rec}} + \lambda_{\mathrm{prior}}\mathcal{L}_{\mathrm{prior}} \\ + \lambda_{\mathrm{freq}}\mathcal{L}_{\mathrm{freq}} + \lambda_{\mathrm{margin}}\mathcal{L}_{\mathrm{margin}}, \end{aligned} \tag{6}$$

where

$$\begin{aligned} \mathcal{L}_{\mathrm{rec}} &= \|\hat{x}_{m_t} - x_{m_t}\|_1, \\ \mathcal{L}_{\mathrm{prior}} &= \|p_0 - x_{m_t}\|_1, \\ \mathcal{L}_{\mathrm{freq}} &= \|\mathcal{W}(\hat{x}_{m_t}) - \mathcal{W}(x_{m_t})\|_1, \\ \mathcal{L}_{\mathrm{margin}} &= \max(0, \mathcal{L}_{\mathrm{rec}} - \mathcal{L}_{\mathrm{prior}} + \delta). \end{aligned} \tag{7}$$

Here, $\mathcal{W}(\cdot)$ is the Haar wavelet transform, and $\delta \geq 0$ specifies the desired reconstruction-error reduction after refinement. The objective supervises both the anatomical prior and the refined output. Specifically, $\mathcal{L}_{\mathrm{prior}}$ provides direct image-level supervision for the initialization, while $\mathcal{L}_{\mathrm{rec}}$ promotes agreement between the synthesized image and the target intensities. The frequency term complements these reconstruction losses by penalizing discrepancies in low- and high-frequency wavelet coefficients.

The margin term explicitly relates reconstruction quality before and after refinement. It becomes active when $\mathcal{L}_{\mathrm{rec}} > \mathcal{L}_{\mathrm{prior}} - \delta$ and vanishes once the refined output achieves the prescribed improvement over the prior. This term therefore links the refinement objective to the quality of its starting point. Together with $\mathcal{L}_{\mathrm{FM}}$, these losses combine velocity supervision along the interpolation path with image- and frequency-domain supervision of the reconstructed target.

TABLE I
DATASET SPLITS. SLICE COUNTS REFER TO AXIAL SLICES USED AS 2.5D SAMPLES.

Dataset	Split	Modalities	Subj.	Slices
BraTS2021	Train	T1/T1CE/T2/FLAIR	1000	155,000
	Val	T1/T1CE/T2/FLAIR	251	38,905
CHAOS	Train	T1in/T1out	32	1,014
	Val	T1in/T1out	8	286
Diff5T	Train	T1/T2	40	10,290
	Test	T1/T2	10	2,210

E. Inference

Given $\mathcal{S}$ and m_t, the fusion encoder and prior generator produce p_0 and Γ_t. We initialize $z^{(0)} = p_0$ and integrate from $\lambda = 0$ to 1 using N Euler steps:

$$z^{(i+1)} = z^{(i)} + \Delta\lambda\, v_\theta(z^{(i)}, \lambda_i; \Gamma_t), \tag{8}$$

where $i = 0, \ldots, N-1$, $\Delta\lambda = 1/N$, and $\lambda_i = i\Delta\lambda$. The final output is $\hat{x}_{m_t} = z^{(N)}$. Inference is deterministic and requires no random-noise initialization.

III. Experiments

A. Datasets and Preprocessing

We evaluate 32 synthesis tasks across three datasets (Table I). BraTS2021 [29] contains co-registered T1, T1CE, T2, and FLAIR scans and defines 28 tasks: 12 single-source, 12 two-source, and 4 three-source tasks. CHAOS [30] provides abdominal T1 in-phase (T1in) and out-of-phase (T1out) scans, while Diff5T [31] provides paired 5T brain T1/T2 scans;

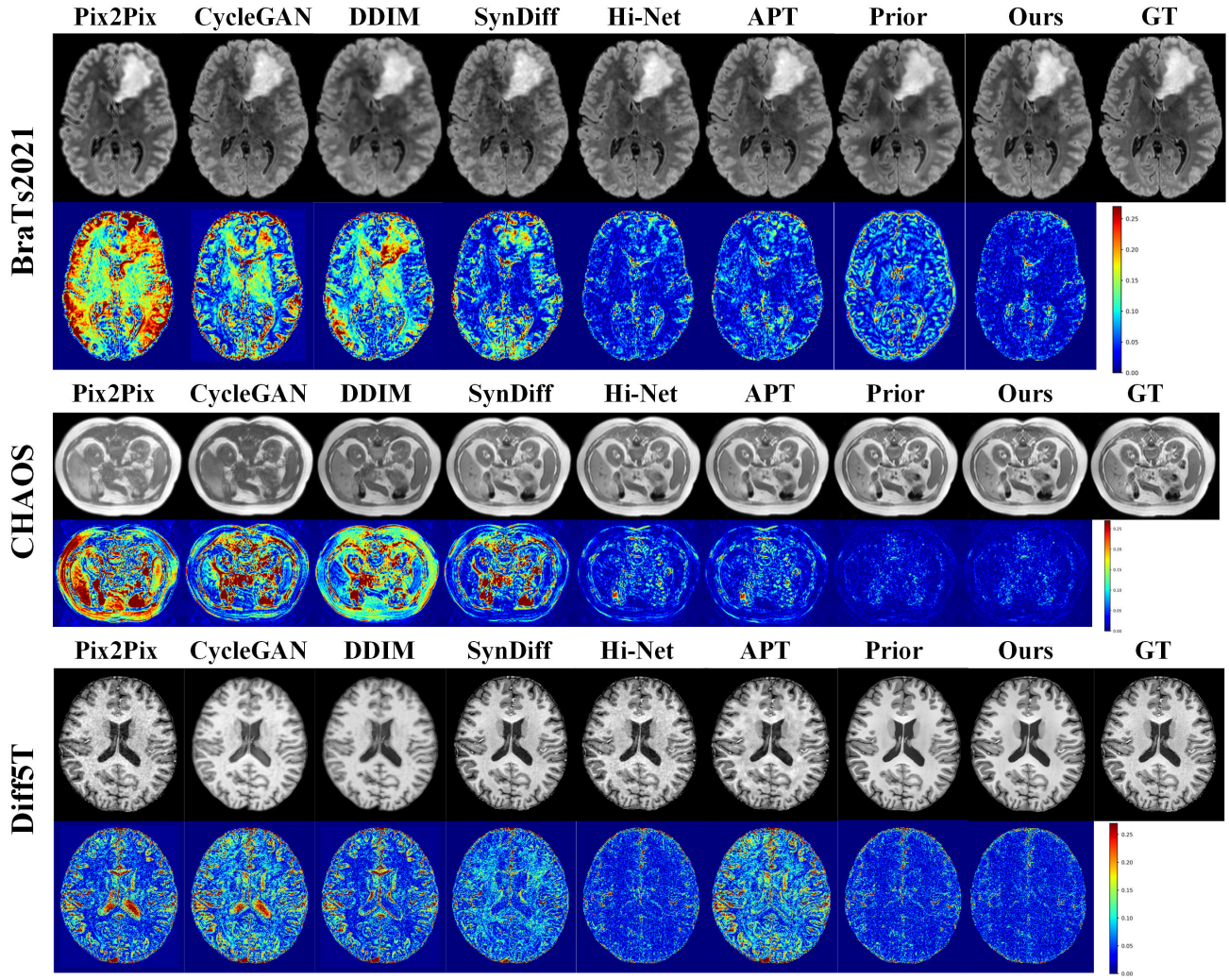


Fig. 3. Qualitative synthesis results on BraTS2021, CHAOS, and Diff5T. For each dataset, the upper row shows target images, including the coarse prior and ground truth (GT); the lower row shows absolute errors relative to GT.

each defines two synthesis directions. BraTS2021 and CHAOS use subject-level five-fold protocols, and Diff5T uses a fixed training/test split.

All volumes are processed as axial 2.5D samples. Foreground intensities are clipped to the $[0.5, 99.5]$ percentiles and min–max normalized to $[0, 1]$ per modality. Without resampling, slices are center-padded to 192×224, 400×400, and 256×288 for BraTS2021, CHAOS, and Diff5T, respectively.

B. Implementation Details

We train one PriorFM model per dataset, sharing parameters across its source–target configurations. The PyTorch implementation uses distributed data-parallel training on A800 and RTX 5090 GPUs. PriorFM is trained for 100 epochs with AdamW, an initial learning rate and weight decay of 10^{-4}, cosine learning-rate decay with warmup, bf16 mixed precision, gradient clipping at norm 1.0, and random seed 42. The loss weights $(\lambda_{\mathrm{FM}}, \lambda_{\mathrm{rec}}, \lambda_{\mathrm{prior}}, \lambda_{\mathrm{freq}}, \lambda_{\mathrm{margin}})$ are $(1.0, 0.5, 0.5, 0.1, 0.1)$. Inference uses 10 Euler steps.

Diff5T additionally uses a discrete latent prior from a frozen pretrained tokenizer with 256 tokens and 14-bit codes. For BraTS2021 and CHAOS, the checkpoint with the highest validation PSNR is evaluated on the corresponding validation split. For Diff5T, the final checkpoint is evaluated on the test split.

C. Results

We compare PriorFM with Pix2Pix [8], CycleGAN [9], DDIM [14], SynDiff [15], Hi-Net [20], and APT [21], using the same preprocessing, subject splits, input resolutions, and metrics. Pairwise methods are evaluated on one-to-one tasks; multi-source-capable methods additionally support two-to-one and three-to-one protocols. PriorFM covers all configurations within each dataset without task-specific retraining.

1) Synthesis: We report peak signal-to-noise ratio (PSNR), structural similarity index (SSIM), and metrically averaged mean squared error (MSE). PSNR and SSIM are averaged over tasks within each protocol or dataset. On BraTS2021,

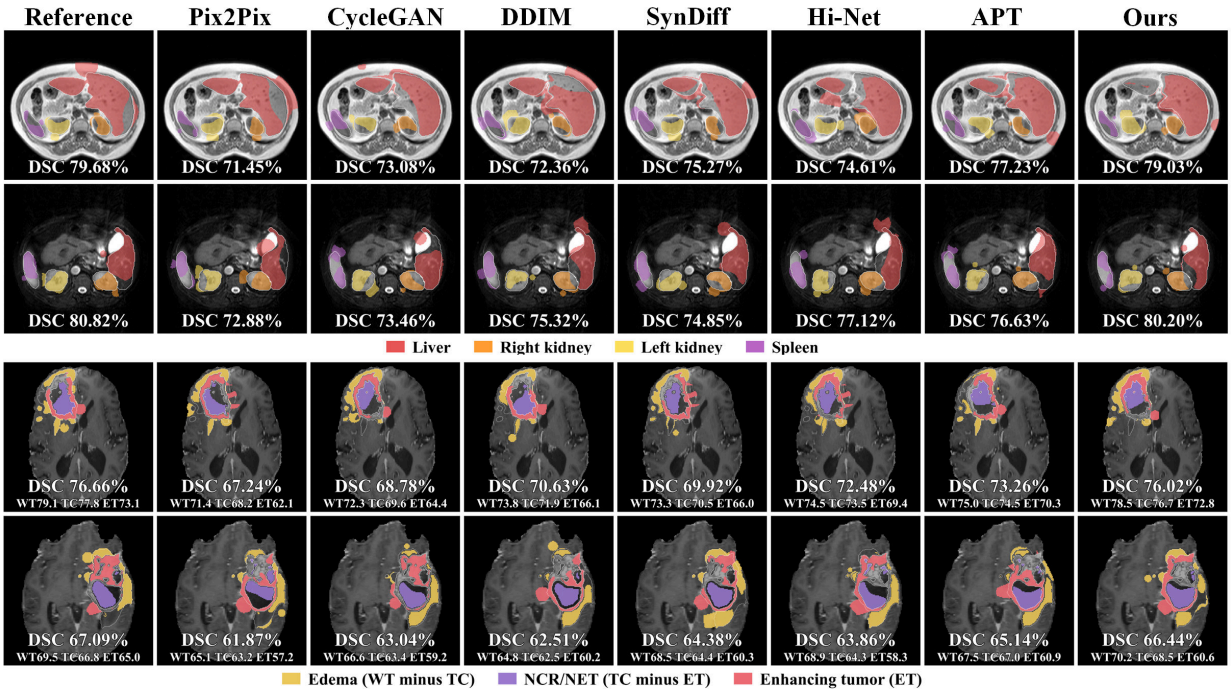


Fig. 4. SegMamba predictions on CHAOS (top two rows) and BraTS2021 (bottom two). Reference uses real images; other columns use synthesized modalities. Colors identify organs and tumor regions as indicated. DSC is shown per example; WT, TC, and ET denote whole tumor, tumor core, and enhancing tumor.

TABLE II
BRATS2021 SYNTHESIS, AVERAGED OVER SOURCE SUBSETS FOR EACH TARGET AND PROTOCOL. PSNR: DB; SSIM: %; MSE: $\times 10^2$. BEST RESULTS ARE BOLD; PRIORFM IS SHADED.

Protocol	Method	T1			T1CE			T2			FLAIR		
		PSNR ↑	SSIM ↑	MSE ↓	PSNR ↑	SSIM ↑	MSE ↓	PSNR ↑	SSIM ↑	MSE ↓	PSNR ↑	SSIM ↑	MSE ↓
1-to-1	Pix2Pix [8]	23.35	84.34	0.462	24.10	81.97	0.389	23.80	82.73	0.417	23.73	82.33	0.423
	CycleGAN [9]	23.92	83.14	0.406	25.13	80.48	0.307	24.54	88.53	0.351	23.07	82.28	0.494
	DDIM [14]	25.78	87.86	0.264	28.21	88.95	0.151	27.52	89.55	0.177	27.08	82.55	0.196
	SynDiff [15]	25.59	90.31	0.276	26.70	83.59	0.214	26.55	85.73	0.221	25.72	87.04	0.268
	Hi-Net [20]	27.33	91.58	0.185	26.60	87.17	0.219	25.99	87.71	0.252	26.39	91.49	0.230
	APT [21]	26.01	88.27	0.251	28.12	87.15	0.154	27.01	89.08	0.199	26.58	86.54	0.220
	Ours	**28.06**	**94.64**	**0.156**	**28.25**	**92.13**	**0.150**	**28.48**	**93.57**	**0.142**	**28.24**	**92.98**	**0.150**
2-to-1	DDIM	26.03	89.23	0.250	26.51	89.34	0.223	25.96	88.73	0.254	26.16	87.91	0.242
	Hi-Net	27.48	90.47	0.179	26.97	88.53	0.201	27.12	90.76	0.194	26.61	87.12	0.218
	APT	26.07	88.39	0.247	27.56	89.47	0.176	26.70	90.08	0.214	27.20	88.95	0.190
	Ours	**28.62**	**95.33**	**0.138**	**29.11**	**93.35**	**0.123**	**29.26**	**94.81**	**0.119**	**28.75**	**93.95**	**0.133**
3-to-1	DDIM	27.07	85.86	0.196	27.64	87.43	0.172	28.03	88.83	0.158	27.73	86.20	0.169
	APT	25.95	87.40	0.254	27.31	90.40	0.186	28.01	89.30	0.158	27.25	90.26	0.189
	Hi-Net	27.71	89.03	0.169	28.85	91.40	0.130	28.52	90.43	0.140	27.40	90.19	0.182
	Ours	**27.76**	**95.55**	**0.168**	**29.57**	**94.03**	**0.110**	**29.46**	**95.24**	**0.113**	**29.11**	**94.46**	**0.123**

PriorFM achieves the best results across all target modalities and source-count protocols (Table II). Averaging the four target-wise PSNR values gives 28.26, 28.94, and 28.98 dB for the one-to-one, two-to-one, and three-to-one protocols, respectively. These results indicate that the shared model can exploit complementary information from additional observed modalities. On CHAOS and Diff5T, PriorFM improves average PSNR over the strongest competing method by 2.70 and 4.50 dB, respectively (Table IV). Figure 3 illustrates anatomical detail and reduced localized errors after prior-to-target refinement.

2) Segmentation: We use SegMamba [32] and MedSAM2 [33] to evaluate segmentation on synthesized BraTS2021 and CHAOS modalities [34], with identical settings across synthesis methods. Dice similarity coefficient (DSC) and intersection-over-union (IoU) measure region overlap; the 95th-percentile Hausdorff distance (HD95), in mil-

TABLE III
SEGMENTATION ON REAL (GT) AND SYNTHESIZED IMAGES. EACH CELL REPORTS SEGMAMBA [32]/MEDSAM2 [33]. DSC/IOU: %; HD95: MM. BEST SYNTHESIS RESULTS ARE BOLD; PRIORFM IS SHADED.

Dataset	Metric	GT	Pix2Pix	CycleGAN	DDIM	SynDiff	Hi-Net	APT	**Ours**
BraTS2021 [29]	DSC (%) ↑	75.66 / 66.09	64.24 / 62.27	68.12 / 62.26	71.43 / 64.77	72.43 / 64.17	73.15 / 63.29	72.68 / 64.27	**75.52 / 65.84**
	IoU (%) ↑	70.24 / 60.71	63.10 / 59.42	65.95 / 60.01	66.21 / 60.52	64.75 / 60.42	66.45 / 60.03	66.12 / 59.44	**69.44 / 60.89**
	HD95 (mm) ↓	3.85 / 15.78	10.35 / 31.23	11.58 / 29.85	9.89 / 27.39	8.87 / 28.69	8.27 / 25.46	9.86 / 23.78	**7.01 / 21.00**
CHAOS [30]	DSC (%) ↑	85.68 / 91.82	78.45 / 84.88	81.05 / 86.90	83.56 / 88.76	82.11 / 87.54	85.12 / 90.88	84.23 / 89.95	**86.03 / 91.50**
	IoU (%) ↑	77.03 / 85.03	72.88 / 78.43	73.56 / 79.55	74.93 / 81.12	75.12 / 81.45	75.88 / 82.97	75.91 / 82.56	**77.34 / 84.50**
	HD95 (mm) ↓	5.89 / 1.80	10.46 / 5.23	10.88 / 6.85	9.12 / 4.49	8.79 / 5.69	7.65 / 3.35	8.53 / 3.48	**6.69 / 1.97**

TABLE IV
SYNTHESIS ON CHAOS AND DIFF5T, AVERAGED OVER BOTH DIRECTIONS. UNITS AND HIGHLIGHTING FOLLOW TABLE II.

Dataset	Metric	Pix2Pix	CycleGAN	DDIM	SynDiff	Hi-Net	APT	**Ours**
CHAOS	PSNR ↑	28.15	28.67	30.01	29.68	33.65	30.75	**36.35**
	SSIM ↑	87.34	88.95	90.12	88.31	90.66	89.78	**92.93**
	MSE ↓	0.1531	0.1358	0.0998	0.1076	0.0432	0.0841	**0.0232**
Diff5T	PSNR ↑	27.90	28.51	29.85	30.59	31.53	31.91	**36.41**
	SSIM ↑	86.93	88.14	88.77	87.79	88.60	88.81	**90.15**
	MSE ↓	0.1622	0.1409	0.1035	0.0873	0.0703	0.0644	**0.0229**

TABLE V
COMPONENT ABLATION ON BRATS2021. A CHECKMARK ENABLES THE COMPONENT. UNITS FOLLOW TABLE II; THE FULL MODEL IS SHADED.

p_0 init.	Haar	Task fusion	Prior ctx.	PSNR ↑	SSIM ↑	MSE ↓
✓				27.43	92.99	0.181
✓	✓			28.37	93.48	0.146
✓		✓		28.21	93.57	0.151
✓			✓	28.35	93.55	0.146
✓	✓	✓	✓	**28.65**	**93.98**	**0.136**

TABLE VI
EULER-STEP ABLATION ON BRATS2021. MSE USES THE NORMALIZED RANGE $[0, 1]$. BEST RESULTS ARE BOLD; THE DEFAULT IS SHADED.

Steps	PSNR ↑	SSIM (%) ↑	MSE ($\times 10^2$) ↓
1	28.21	93.63	0.151
2	28.35	93.69	0.146
4	28.48	93.75	0.142
8	28.63	93.88	0.137
10	**28.65**	**93.98**	**0.136**
16	28.62	93.86	0.137

limeters, measures boundary agreement. Scores are averaged over target modalities on BraTS2021 and both synthesis directions on CHAOS. Segmentation on real images (GT) provides a reference outside the synthesis-method ranking.

PriorFM achieves the highest DSC/IoU and lowest HD95 among synthesis methods on both datasets and backbones (Table III). With SegMamba, BraTS2021 DSC/HD95 are $75.52\%/7.01$ mm for PriorFM and $75.66\%/3.85$ mm for real images; CHAOS scores are $86.03\%/6.69$ mm and $85.68\%/5.89$ mm, respectively. Figure 4 shows representative organ and tumor predictions.

D. Ablation Study

Both ablations report averages over all BraTS2021 tasks.

1) Components: The baseline initializes flow refinement from p_0 and achieves 27.43 dB PSNR. Adding Haar frequency cues, task-aware fusion, or multi-scale prior context individually raises PSNR to 28.37, 28.21, and 28.35 dB, respectively (Table V). These components provide frequency information, target-dependent source aggregation, and multi-scale anatomical conditioning. Their combination achieves the highest PSNR (28.65 dB) and SSIM (93.98%), improving over the baseline by 1.22 dB and 0.99 percentage points. It also exceeds the strongest individual addition by 0.28 dB, supporting their complementary contributions.

2) Sampling Steps: Table VI evaluates the effect of Euler steps on synthesis quality. A single step achieves 28.21 dB PSNR and 93.63% SSIM. PSNR rises to 28.48 dB at four steps and 28.63 dB at eight steps, with larger gains during the initial updates. Ten steps yield the highest PSNR (28.65 dB) and SSIM (93.98%), improving upon single-step inference by 0.44 dB and 0.35 percentage points, respectively. Increasing the step count to sixteen provides no further improvement. We therefore adopt ten Euler steps as the default configuration.

IV. CONCLUSIONS

PriorFM provides a unified framework for missing-modality MRI synthesis, using one task-conditioned model per dataset for varying source subsets and target modalities. It combines task-aware fusion, an anatomical prior, and frequency-guided conditional flow refinement. Across 32 tasks on BraTS2021, CHAOS, and Diff5T, PriorFM improves synthesis quality in single-source and multi-source settings with deterministic inference using ten Euler steps. Evaluations with SegMamba and MedSAM2 yield Dice scores close to those obtained from real images, supporting downstream segmentation utility. Component ablations support the complementary contributions of task-aware fusion, frequency guidance, and prior-context conditioning. These results support prior-to-target transport for flexible MRI modality completion. Future work will explore full 3D synthesis and broader modality sets.

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